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B. E. (Sixth Semester) Examination, Nov.-Dec. 2015

(Old Scheme)

(Elect., EEE Engg.)

DIGITAL SIGNAL PROCESSING

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Attempt all questions. Part (a) of each question is compulsory & carries 2 marks. Solve any two parts from part (b), (c) and (d) & carries 7 marks each.

Unit-I

1. (a) Compute :

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$$\int_{-\infty}^{\infty} \left(t^2 + 2t + 5\right) s(t)\, dt$$

(b) Explain the different way to represent signals & systems.

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(c) Sketch the single-sided and double-sided amplitude and phase spectra of the signal :

$$x(t) = 8 \sin\left(20\pi t - \frac{\pi}{6}\right) \quad -\infty < t < \infty$$

(d) With a block diagram explain the process of analog to digital conversion.

Unit-II

2. (a) Define Causality.

(b) Using Paley Wiener interior, show that $|H(jw)| = e^{-e^{2}}$ is not a suitable amplitude response for a causal LTI system.

(c) Digital filter is characterised by the difference equation :

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$Y(n) = x(n) + e^{a}\, y(n-1)$ check the filters for BIBO stability.

(d) Determine the magnitude response and frequency response of the system given by :

$$y(n) + \tfrac{1}{2} Y(n-1) = x(n) - x(n-1)$$

Unit-III

3. (a) What do you mean by Impulse Response?

(b) Explain the difference between Linear Convolution and Circular Convolution.

(c) Find the response of an FIR filter with impulse response $h(n) = \{1, 2, 4\}$ to the input sequence $x(n) = \{1, 2\}$.

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(d) Compute the DFTs of the sequence $x(n) = \cos\frac{n\pi}{2}$ where $N = 4$ using DIFFFT.

Unit-IV

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4. (a) What is Frequency warping?

(b) Use the backward difference for the derivative and convert the analog filter with system function :

$$H(s) = \frac{1}{s^2 + 16}$$

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(c) A digital filter with 3 db bandwidth of $0 \cdot 25\ \pi$ is to be designed from the analog filter whose system response is :

$$H(s) = \frac{\Omega_c}{s + \Omega_c}$$

Use Bilinear transformation & obtain $H(z)$.

(d) Describe Eliptic Filters.

Unit-V

5. (a) What do you mean by Canonic structure?

(b) Determine Direct form-I and Direct form-II for the second order filter given by :

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$$y(n) = 2b \cos w_0\, y(n+1) - b^2\, y(n-2) + x(n)$$

$$- b \cos w_0\, x(n-1)$$

(c) Determine the parallel realisation of the IIR digital filter transfer function :

$$H(z) = \frac{3\left(2z^2 + 5z + 4\right)}{(2z+1)(z+2)}$$

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(d) Give comparision of Fourier transform and wavelet transform.

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